E0 297: Cryptographic Hardware Accelerators

Paper Review Assignment

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Aug 2026



Objectives

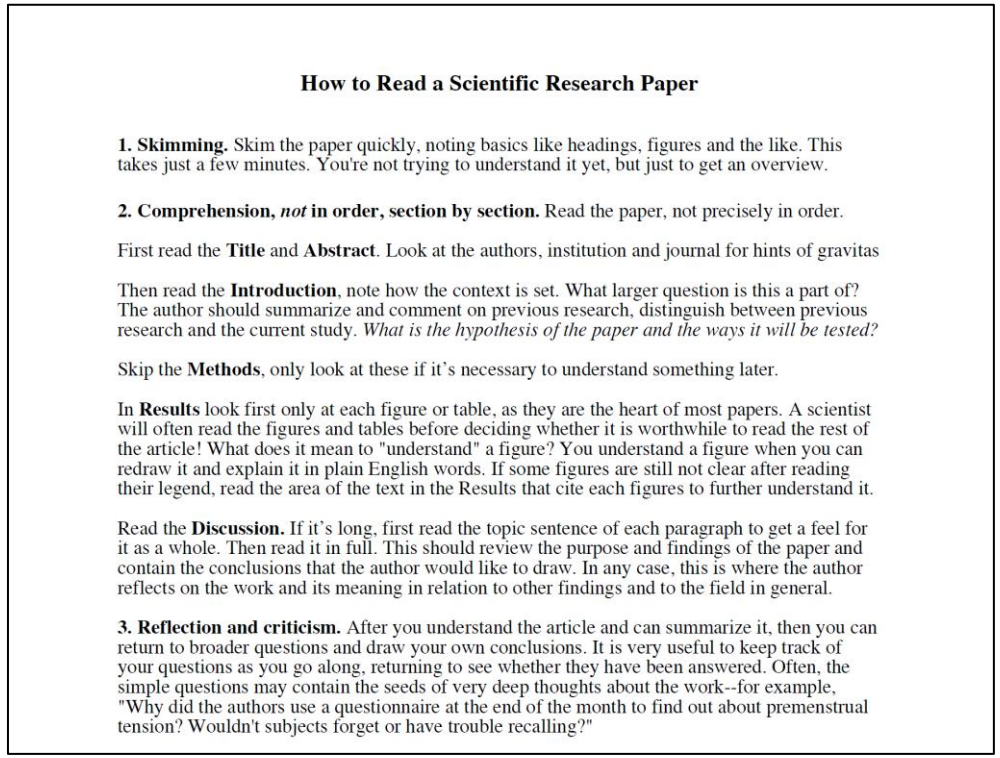
- Study research papers on topics discussed in the lectures
- Learn how to read papers and critically analyse their claims
- Understand how peer review works in real life conferences / journals
- Participate in group discussion to reach consensus on peer reviews
- Get potential ideas for final project



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Pre-Requisite

- Carefully go through the **following guidelines on effectively reading papers** (both documents uploaded in the course Teams group under **General > Shared > Class Materials > Reference Materials**)



How to Read a Scientific Research Paper

1. Skimming. Skim the paper quickly, noting basics like headings, figures and the like. This takes just a few minutes. You're not trying to understand it yet, but just to get an overview.

2. Comprehension, *not* in order, section by section. Read the paper, not precisely in order.

First read the **Title** and **Abstract**. Look at the authors, institution and journal for hints of gravitas

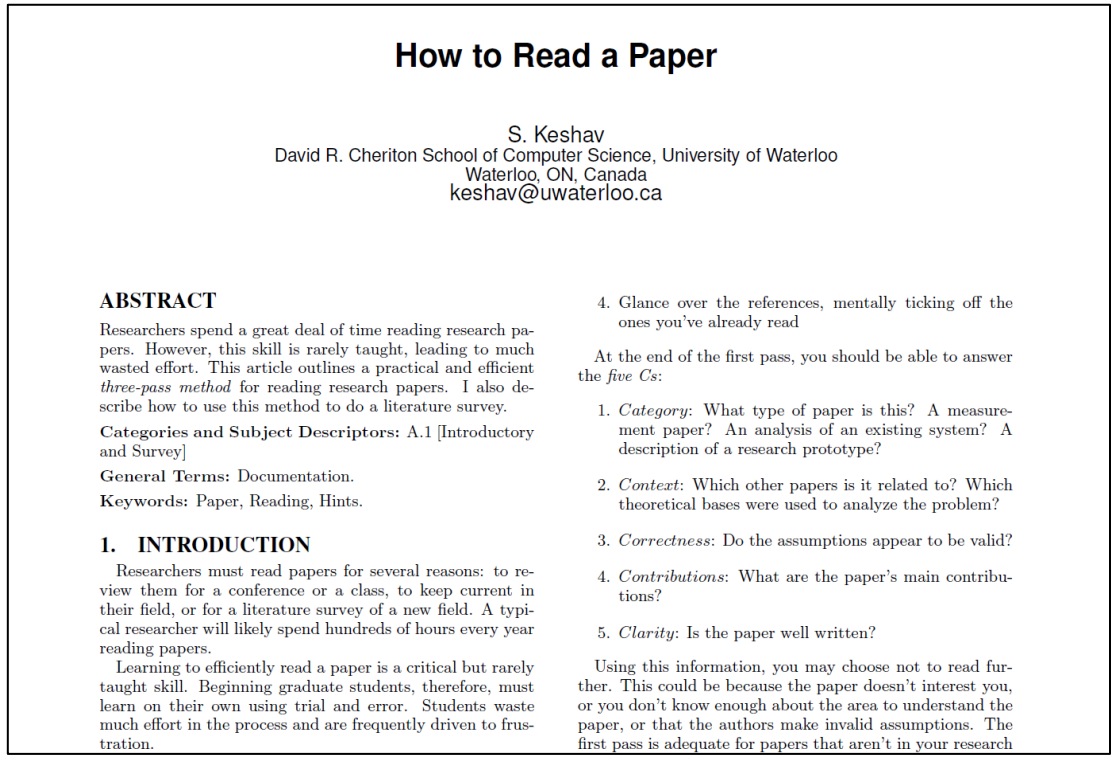
Then read the **Introduction**, note how the context is set. What larger question is this a part of? The author should summarize and comment on previous research, distinguish between previous research and the current study. *What is the hypothesis of the paper and the ways it will be tested?*

Skip the **Methods**, only look at these if it's necessary to understand something later.

In **Results** look first only at each figure or table, as they are the heart of most papers. A scientist will often read the figures and tables before deciding whether it is worthwhile to read the rest of the article! What does it mean to "understand" a figure? You understand a figure when you can redraw it and explain it in plain English words. If some figures are still not clear after reading their legend, read the area of the text in the Results that cite each figures to further understand it.

Read the **Discussion.** If it's long, first read the topic sentence of each paragraph to get a feel for it as a whole. Then read it in full. This should review the purpose and findings of the paper and contain the conclusions that the author would like to draw. In any case, this is where the author reflects on the work and its meaning in relation to other findings and to the field in general.

3. Reflection and criticism. After you understand the article and can summarize it, then you can return to broader questions and draw your own conclusions. It is very useful to keep track of your questions as you go along, returning to see whether they have been answered. Often, the simple questions may contain the seeds of very deep thoughts about the work--for example, "Why did the authors use a questionnaire at the end of the month to find out about premenstrual tension? Wouldn't subjects forget or have trouble recalling?"



How to Read a Paper

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ABSTRACT

Researchers spend a great deal of time reading research papers. However, this skill is rarely taught, leading to much wasted effort. This article outlines a practical and efficient *three-pass method* for reading research papers. I also describe how to use this method to do a literature survey.

Categories and Subject Descriptors: A.1 [Introductory and Survey]

General Terms: Documentation.

Keywords: Paper, Reading, Hints.

1. INTRODUCTION

Researchers must read papers for several reasons: to review them for a conference or a class, to keep current in their field, or for a literature survey of a new field. A typical researcher will likely spend hundreds of hours every year reading papers.

Learning to efficiently read a paper is a critical but rarely taught skill. Beginning graduate students, therefore, must learn on their own using trial and error. Students waste much effort in the process and are frequently driven to frustration.

4. Glance over the references, mentally ticking off the ones you've already read

At the end of the first pass, you should be able to answer the *five Cs*:

1. *Category*: What type of paper is this? A measurement paper? An analysis of an existing system? A description of a research prototype?
2. *Context*: Which other papers is it related to? Which theoretical bases were used to analyze the problem?
3. *Correctness*: Do the assumptions appear to be valid?
4. *Contributions*: What are the paper's main contributions?
5. *Clarity*: Is the paper well written?

Using this information, you may choose not to read further. This could be because the paper doesn't interest you, or you don't know enough about the area to understand the paper, or that the authors make invalid assumptions. The first pass is adequate for papers that aren't in your research

Paper Review Tracks

- Papers grouped into the following 4 tracks:
  1. AES Hardware Accelerators (3 papers)
  2. SHA2 / SHA3 Hardware Accelerators (3 papers)
  3. ASCON Hardware Accelerators (3 papers)
  4. Physical-Attack-Resilient Crypto Building Blocks (3 papers)
- Track preferences collected through paper review preference survey
- Expect random assignment if preference survey not submitted

Track 1 Papers

Track 1 - AES Hardware Accelerators:

- **Paper #1:** P. Hamalainen et al., “Design and Implementation of Low-Area and Low-Power AES Encryption Hardware Core,” EuroMicro Conference on Digital System Design, 2006: https://ieeexplore.ieee.org/abstract/document/1690090
- **Paper #2:** B. Erbagci et al., “A 275 Gbps AES Encryption Accelerator using ROM-based S-boxes in 65nm,” IEEE Custom Integrated Circuits Conference, 2015: https://ieeexplore.ieee.org/abstract/document/7338448
- **Paper #3:** C. Duran et al., “A 10pJ/bit 256b AES-SoC Exploiting Memory Access Acceleration,” IEEE Transactions on Circuits and Systems II: Express Briefs, 2022: https://ieeexplore.ieee.org/abstract/document/9610111

Track 2 Papers

Track 2 - SHA2 / SHA3 Hardware Accelerators:

- **Paper #1:** R. Ramanarayanan et al, “18Gbps, 50mW Reconfigurable Multi-Mode SHA Hashing Accelerator in 45nm CMOS,” IEEE European Solid-State Circuits Conference, 2010: https://ieeexplore.ieee.org/abstract/document/5619892
- **Paper #2:** M. M. Wong et al., “A New High Throughput and Area Efficient SHA-3 Implementation,” IEEE International Symposium on Circuits and Systems, 2018: https://ieeexplore.ieee.org/abstract/document/8351649
- **Paper #3:** K. Nagarajan et al., “SHINE: A Novel SHA-3 Implementation Using ReRAM-based In-Memory Computing,” IEEE/ACM International Symposium on Low Power Electronics and Design, 2019: https://ieeexplore.ieee.org/abstract/document/8824979

Track 3 Papers

Track 3 - ASCON Hardware Accelerators:

- **Paper #1:** H. Gross et al., “Suit up! -- Made-to-Measure Hardware Implementations of ASCON,” EuroMicro Conference on Digital System Design, 2015: https://ieeexplore.ieee.org/abstract/document/7302339
- **Paper #2:** K.-D. Nguyen et al., “ASIC Implementation of ASCON Lightweight Cryptography for IoT Applications,” IEEE Transactions on Circuits and Systems II: Express Briefs, 2025: https://ieeexplore.ieee.org/abstract/document/10722846
- **Paper #3:** M. Lee et al., “A Compute-in-Memory Ascon Implementation Based on a Novel 11T SRAM Processing Macro,” IEEE Symposium on Low-Power and High-Speed Chips and Systems, 2025: https://ieeexplore.ieee.org/abstract/document/11018599

Track 4 Papers

Track 4 - Physical-Attack-Resilient Crypto Building Blocks:

- **Paper #1:** K. Tiri et al., "A Dynamic and Differential CMOS Logic with Signal Independent Power Consumption to Withstand Differential Power Analysis on Smart Cards," IEEE European Solid-State Circuits Conference, 2002: https://ieeexplore.ieee.org/abstract/document/1471550
- **Paper #2:** K. Tiri et al., "A Logic Level Design Methodology for a Secure DPA Resistant ASIC or FPGA Implementation," Design, Automation and Test in Europe, 2004: https://ieeexplore.ieee.org/abstract/document/1268856
- **Paper #3:** S. Lu et al., "A 1.25pJ/bit 0.048mm^2 AES core with DPA Resistance for IoT Devices," IEEE Asian Solid-State Circuits Conference, 2017: https://ieeexplore.ieee.org/abstract/document/8240217

Final Deliverables for Paper Review

- **Paper Summary** (succinct overview of the paper in 150-200 words; do not copy directly from paper abstract and/or conclusion; write in your words based on your understanding and analysis of the paper)
- **Strengths** (3 strengths of the research; include 2-3 sentences for each point to justify why you think it is a strength / novelty)
- **Weaknesses** (3 weaknesses of the research; include 2-3 sentences for each point to justify why you think it is a weakness / shortcoming)
- **Reviewer Confidence** (Low / Medium / High)
- **Overall Rating of the Paper** (Average / Good / Very Good)

Paper Review Rules

- Use of AI is strictly prohibited for this paper review assignment
- You are not allowed to discuss with others reviewing the same paper
- You must read the paper thoroughly and write the paper review by yourself based on your own understanding
- This policy is in line with existing guidelines from IEEE and ACM regarding peer reviews in real life conferences / journals

References

- **Required:**
  - S. Keshav, “How to Read a Paper,” ACM SIGCOMM Computer Communication Review, 2007: https://dl.acm.org/doi/pdf/10.1145/1273445.1273458